

Quiz 9 (08 November 2024)

In each question explain how you arrive at your answer.

Consider the natural basis $S = \{1, \xi\}$ of the field $K = \mathbb{Q}[\xi]$ where ξ is a root of the polynomial $4X^2 + 2X + 1$.

Question 1

This defines an injective map M_S from K to the ring $M_2(\mathbb{Q})$ of 2×2 matrices as explained in class.

Calculate the order $R_1 = \{\alpha \in K : M_S(\alpha) \in M_2(\mathbb{Z})\}$ by giving an *integral* basis.

Solution. Note that $M_S(1)$ is the identity matrix and

$$M_S(\xi) = \begin{pmatrix} 0 & 1 \\ -\frac{1}{4} & -\frac{1}{2} \end{pmatrix}$$

Thus R_1 contains 1 and 4ξ . Moreover, for rational numbers a and b , we have

$$M_S(a + b\xi) = \begin{pmatrix} a & b \\ -\frac{b}{4} & \frac{2a-b}{2} \end{pmatrix}$$

This is an integer matrix if and only if a is an integer and b is divisible by 4. It follows that the required basis is 1 and 4ξ .

Question 2

Note that $1/\xi$ is an algebraic integer.

Consider the order $R = \mathbb{Z} + \mathbb{Z} \cdot (1/\xi)$ in K . Calculate

$$R^* = \{\alpha \in K : \text{Tr}_{K/\mathbb{Q}}(\alpha\beta) \in \mathbb{Z} \text{ for all } \beta \in R\}$$

by giving an integral basis of it.

Recall that $\text{Tr}_{K/\mathbb{Q}}(\alpha)$ is the trace of the matrix $M_S(\alpha)$ for α in K .

Solution. Note that $1/\xi = -2 - 4\xi$ by using the given equation.

$$(a + b\xi)(c + d/\xi) = (ac + bd) + bc\xi + ad(-2 - 4\xi) = 2(ac + bd - 2ad) + (bc - 4ad)\xi$$

Note that by the matrices calculated above

$$\begin{aligned} \text{Tr}_{K/\mathbb{Q}}(1) &= 2 \\ \text{Tr}_{K/\mathbb{Q}}(\xi) &= -1/2 \end{aligned}$$

Thus, if $\alpha = a + b\xi$ and $\beta = c + d/\xi$, then

$$\text{Tr}_{K/\mathbb{Q}}(\alpha\beta) = 2(ac + bd - 2ad) - \frac{bc - 4ad}{2}$$

$$\begin{aligned}\mathrm{Tr}_{K/\mathbb{Q}}(\alpha \cdot 1) &= 2a - b/2 \\ \mathrm{Tr}_{K/\mathbb{Q}}(\alpha \cdot (1/\xi)) &= 2(b - 2a) + 2a = 2b - 2a\end{aligned}$$

Thus, it is necessary and sufficient that $2a - b/2 = p$ and $2(b - a) = q$ are integers.

This gives $b = 2p/3 + 2q/3$ and $a = 2p/3 + q/6$.

It follows that $(2/3)(1 + \xi)$ and $1/6 + (2/3)\xi$ give a basis for R^* .